(a) Fixed incident beam

$E_{in}(x, y)$: spatial amplitude and phase

8.2 keV; FWHM = 7.11 μm

Sample 0.25 m downstream of the beam waist

(b) Smooth and translate the wire

Cu cylinder: a = 5 μm; axis parallel to y

Independent widths: $\sigma_L = 4.67$ nm, $\sigma_R = 0$ nm

$t_n(x, y) = t_{\sigma L,\sigma R}[x - x_0^{(n)}]$

43 offsets: −10.0 to +10.16 μm; step 0.48 μm

(c) Construct the sample-exit field

Interpolate attenuation and phase onto the beam grid

$$E_{exit}^{(n)}(x, y) = E_{in}(x, y)\exp[-\mu t_n(x, y)/2]\exp[-ik\delta t_n(x, y)]$$

(d) Fresnel propagation and its Fraunhofer far-field approximation

Sample-plane quadratic phase

$$U_n = E_{exit}^{(n)}\exp[ik(x^2 + y^2)/(2Z)]$$

→ **2D FFT** $\tilde{U}_n(q_x, q_y)$ →

Detector intensity

$$I_D^{(n)} \propto |\tilde{U}_n(kX/Z, kY/Z)|^2$$

Z = 1.31 m; 4096 × 4096 grid; $\Delta x = \Delta y$ = 8 nm; $\Delta X = \Delta Y \approx$ 6.04 μm

Fraunhofer far-field approximation: neglect the sample-plane quadratic phase

$$\exp[ik(x^2 + y^2)/(2Z)] \approx 1 \Rightarrow U_n \approx E_{exit}^{(n)};\ I_D^{(n)} \propto |F\{E_{exit}^{(n)}\}(kX/Z, kY/Z)|^2$$

(e) Extract the SAXS profile and quantify asymmetry

Average 10 central rows: $\bar{I}_n(q_x)$

Use $q_x = kX/Z$; exclude $|q_x| \le q_{min}$

$q_{min} = 0.10\ \text{nm}^{-1}$

LEFT	EXCLUDED	RIGHT

→

Integrate left and right regions

$$S_L = \Sigma_{left}\bar{I}_n,\ S_R = \Sigma_{right}\bar{I}_n$$

$$f_L^{(n)} = \frac{S_L^{(n)}}{S_L^{(n)} + S_R^{(n)}}$$

Repeat for every wire offset